EE150 Signal and System
Homework 6

Note:

- Please provide enough calculation process to get full marks.
- Please submit your homework to Gradescope in PDF version.
- It's highly recommended to write every exercise on a single sheet of page.
- Late submissions will have points deducted according to the penalty policy.
- Please use English only to complete the assignment, solutions in Chinese are deducted according to the penalty policy.
- Plagiarizer will get zero points.
- The full score of this assignment is 100 points.

Exercise 1. (10pt)

A causal LTI filter has the frequency response $H(j\omega)$ shown in Figure 1.

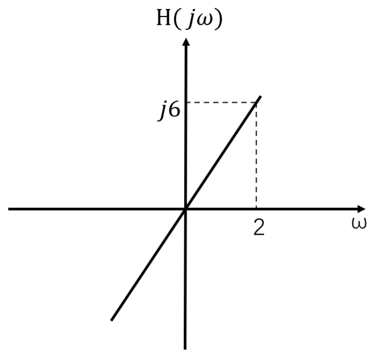

Figure 1

For each of the input signals given below, determine the filtered output signal $y(t)$.

(a) $x(t) = e^{j3t}$

(b) $x(t) = (\cos 2t)u(t)$

(c) $X(j\omega) = \frac{1}{3+j\omega}$

Exercise 2. (20pt)

Consider the stable LTI system

$$\tau \frac{dy(t)}{dt} + y(t) = x(t), \qquad \tau = 0.01 \text{ s}.$$

(a) Derive the frequency response $H(j\omega) = \dfrac{Y(j\omega)}{X(j\omega)}$.

(b) Obtain formulas for $|H(j\omega)|$ and $\angle H(j\omega)$. Determine whether this system is linear-phase.

(c) Derive the expressions for $|H(j\omega)|$ and $\angle H(j\omega)$ with asymptotic approximation, and sketch the corresponding asymptotic Bode plots on the axes in Figure 2.1.

(d) Now consider an input signal

$$x(t) = \big[1 + 0.5\cos(10\pi t)\big]\cos(100\pi t),$$

which is fed into the system with the frequency response $H(j\omega)$. Determine the frequency at which the most of the signal energy concentrates. Then calculate the delay experienced by the information around such frequency.
(*Recall* : $\frac{d}{dx}\arctan(x) = \frac{1}{1+x^2}$.)

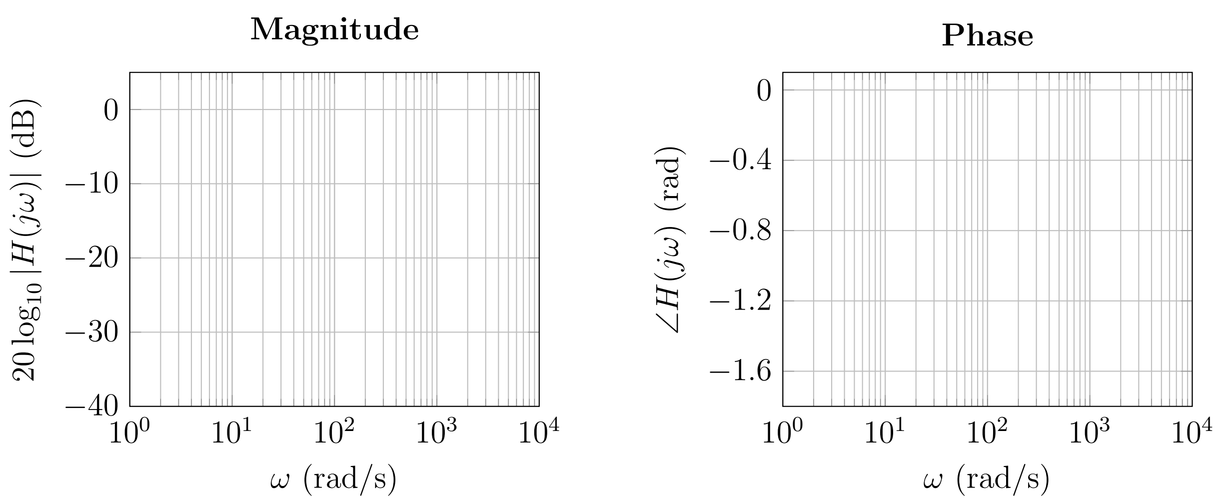

Figure 2.1 Bode magnitude and phase plots of $H(j\omega)$.

Exercise 3. (15pt)

(a) Consider the continuous-time signal

$$x(t) = \cos(1200\pi t)\cos(200\pi t).$$

Determine the condition that the sampling interval T_s must satisfy in order to avoid aliasing.

(b) Consider the continuous-time signal

$$x(t) = \frac{\sin(300\pi t)}{\pi t}.$$

Suppose that $x(t)$ is sampled with $p(t)$ as shown in Figure 3.1 where $T_s = 5$ ms and then we get $x_s(t)$. Sketch the spectrum of $X_(j\omega)$ and $X_s(j\omega)$.

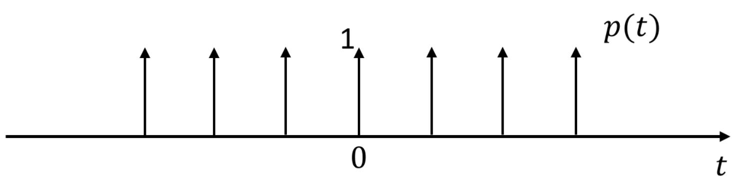

Figure 3.1 Impulse train $p(t)$

Exercise 4. (20pt)

A signal

$$x(t) = \cos(2\pi f_0 t + \theta)$$

is sampled by an impulse train with sampling frequency $f_s = 1000$ Hz. At the receiver, an ideal low-pass filter with cutoff $f_c = 500$ Hz is used to reconstruct the signal.

(a) For $f_0 = 200$ Hz, $\theta = \pi/4$, find the reconstructed signal $x_r(t)$.

(b) For $f_0 = 700$ Hz, $\theta = \pi/2$, find the reconstructed signal $x_r(t)$.

(c) Sketch the spectrum of $x(t)$, $x_r(t)$ from (a) and (b). Then comment on whether aliasing occurs in each case.

Exercise 5. (15pt)

Consider a discrete-time sequence $x[n]$ shown in Figure 5.1

$$n = -7 : 7; x = [0.5, 1, 1.5, 2, 2.5, 2, 1.5, 1, 1.5, 2, 2.5, 2, 1.5, 1, 0.5], x = 0 \text{ for other } n.$$

Now, we perform decimation with a factor of N=2. This process can be summarized as: we first use an impulse train with the sampling interval of N to sample x[n], which yields a discrete-time sequence $x_p[n]$; we then discard the samples obtained at the time when n is not an integer multiples of N, to obtain $x_d[n]$. .

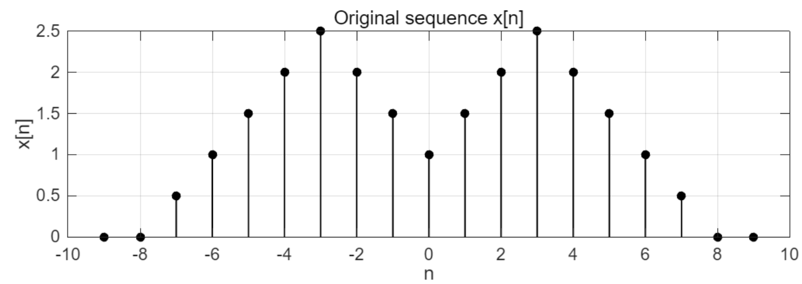

Figure 5.1 $x[n]$

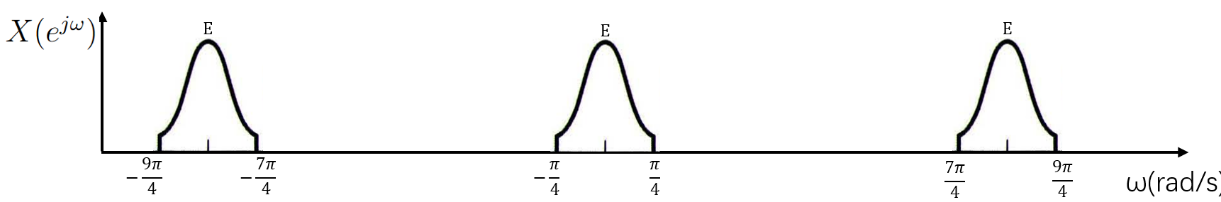

Figure 5.2 $X(e^{j\omega})$.

(a) If $x[n]$ is as illustrated in Figure 5.1, sketch the sequences $x_p[n]$ and $x_d[n]$.

(b) If $X(e^{j\omega})$ is as shown in Figure 5.2, sketch $X_p(e^{j\omega})$ and $X_d(e^{j\omega})$.

Exercise 6. (20 pt)

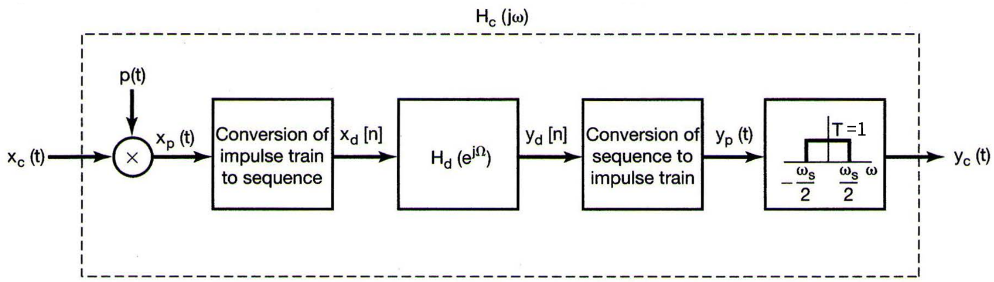

Figure 6.1 Overall system

Consider the overall system in Figure 6.1 for discrete-time processing of continuous-time signals:
The system parameters are:

- Sampling Frequency: $f_s = 2000$ Hz.
- Digital Filter: An ideal low-pass filter

$$H_d(e^{j\Omega}) = \begin{cases} 1, & |\Omega| \leq \frac{3\pi}{4} \\ 0, & \frac{3\pi}{4} < |\Omega| \leq \pi \end{cases}$$

The continuous-time input signal is given by:

$$x_c(t) = 4\cos(2\pi \cdot 500t) + 2\cos(2\pi \cdot 900t)$$

(a) Determine the discrete-time sequence $x_d[n]$. Express the result in the form $x_d[n] = A_1 \cos(\Omega_1 n) + A_2 \cos(\Omega_2 n)$.

(b) Sketch $X_d(e^{j\Omega})$. Find the expression of $y_d[n]$ and sketch its magnitude spectrum.

(c) Determine the effective analog cutoff frequency f_{cutoff} (in Hz) of $H_c(j\omega)$, the entire system.

(d) Find the final reconstructed output signal $y_c(t)$ and its spectrum.